

TSA Meeting Preparation — Detailed Study Note

Slide 13: Ch 9 — Non-Blocking EKF Digital Twin

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Sources: SAMBP TR-43, TR-44, TR-45 / 2026

1. Slide Overview and Purpose

What This Slide Is About

This slide introduces the **Digital Twin (DT)** — the real-time brain of the entire SAMBP framework. While every other slide describes a specific protection element (87L, OC, distance, microgrid), this slide describes the *shared infrastructure* that feeds them all.

The DT continuously estimates three network parameters $\theta = \{k_{ibr}, Z_{line}, Y_{shunt}\}$ from PMU and IED measurements. These estimates are broadcast to every protection module in real time, enabling all adaptive thresholds and confidence gates to respond to the current state of the IBR fleet — without human intervention.

The slide establishes three key properties of the DT: **accuracy** (RMSE results), **speed** (convergence < 2 cycles), and **safety** (non-blocking and bounded update law).

Position in the thesis: Chapter 6 (Monitoring and ML), slide 13 of 22. This is the first of three Ch 6 slides (DT architecture, PINN classifier, online learning). It is a pivotal slide because the DT underpins every contribution: without reliable $\hat{k}_{ibr}$, all adaptive thresholds revert to fixed settings.

2. The Role of the Digital Twin

DT as the Central Parameter Hub

Module receives	What it uses from the DT
87L (Chapter 3)	$\hat{k}_{ibr} \rightarrow$ adaptive slope $k_m^*(k_{ibr})$
87T (Chapter 3)	$\hat{k}_{ibr} \rightarrow$ adaptive transformer bias $k_T^*(k_{ibr})$
SyncOC (Chapter 4)	$\hat{k}_{ibr} \rightarrow$ confidence gate threshold
Distance 21 (Chapter 4)	$\hat{Z}_{line} \rightarrow$ reach correction, IBR shift
Microgrid 67 (Chapter 5)	$\hat{k}_{ibr} \rightarrow$ mode detector prior
PINN (Chapter 6)	$\hat{k}_{ibr} \rightarrow$ feature input to classifier
WAPC (Chapter 7)	all $\theta \rightarrow$ wide-area settings update trigger

Without the DT, every module above would need its own parameter estimator. The DT centralises this into a single, validated, non-blocking component.

3. EKF Prediction–Correction Algorithm

The state vector

The DT tracks three parameters jointly:

- k_{ibr} — IBR penetration ratio (primary, controls all thresholds)
- Z_{line} — line impedance magnitude (for distance reach and fault location)
- Y_{shunt} — shunt admittance (charging current, affects 87L stability)

The update equation (from the slide)

EKF Prediction–Correction (slide equation)

$$\hat{\theta}_{k|k} = \hat{\theta}_{k|k-1} + K_k \left(z_k - h(\hat{\theta}_{k|k-1}) \right) \quad (1)$$

- $\hat{\theta}_{k|k}$ — updated state estimate at step k (posterior)
- $\hat{\theta}_{k|k-1}$ — predicted state (prior, from previous cycle)
- K_k — Kalman gain matrix: $K_k = P_{k|k-1} H_k^\top (H_k P_{k|k-1} H_k^\top + R_k)^{-1}$
- z_k — measurement vector: PMU voltage phasors, IED current measurements, and the sequence current ratio I_2/I_1
- $h(\cdot)$ — nonlinear measurement model (hence *Extended* KF — the Jacobian $H_k = \partial h / \partial \theta$ is evaluated at each step)
- $z_k - h(\hat{\theta}_{k|k-1})$ — innovation: difference between measured and model-predicted outputs; drives the update

Why EKF and not a simple RLS?

Recursive Least Squares (used in TR-43) estimates k_{ibr} alone. The EKF jointly estimates all three states, propagating their *covariance* — it knows how uncertain each estimate is and weighs the Kalman gain accordingly. This matters because:

1. Y_{shunt} and Z_{line} are coupled in the measurement model; estimating them separately introduces bias.
2. The EKF's covariance $P_{k|k}$ feeds directly into the confidence score γ used by every adaptive protection module.
3. Pre-fault warm-starting (tracking θ in normal operation) reduces convergence time by **7×** compared to a cold start.

4. The Non-Blocking Property

Why Non-Blocking Matters — The Safety Argument

If the EKF update were *in the critical path* of the relay trip decision, then:

- A slow convergence or numerical issue in the EKF could **delay a trip signal** by tens of milliseconds — causing a missed fault clearance.
- This would violate IEC 60255 operating time requirements and potentially cause equipment damage or arc flash.

The non-blocking design *guarantees* this cannot happen.

Implementation: the shadow register pattern

Thread	Cycle time	Action
Relay thread (high priority)	50 μ s	Reads $\hat{\theta}$ from the shadow register; executes trip logic; issues GOOSE
EKF thread (low priority)	20 ms (50 Hz)	Runs prediction-correction; writes result to shadow register when done

The relay thread never waits for the EKF thread.

The convergence guard (from the slide)

The slide states: “if convergence > 3 cycles, previous $\hat{\theta}$ is held — relay never waits.”

This means: if the EKF has not converged after 3 update cycles ($3 \times 20 = 60$ ms), the relay continues using the last valid estimate. This is safe because:

- The pre-fault warm-start estimate is already within $\pm 5\%$ of the true value.
- The protection elements are designed with *margin* that accommodates $\pm 10\%$ uncertainty in $\hat{k}_{ibr}$.
- A 3-cycle convergence failure is a diagnostic flag, not a protection failure.

End-to-end latency (TR-45)

Stage	Latency [ms]	Cumulative [ms]
CT/VT transducer	0.5	0.5
IEC 61869-9 digital sampling	0.5	1.0
Merging unit $\rightarrow$ relay	1.0	2.0
Relay algorithm (Z1/87L)	18.0	20.0
GOOSE trip signal	2.0	22.0
Circuit breaker operation	30.0	52.0
Total fault clearance	52.0	
T_{mem} margin	48.0	(within 100 ms)
DT latency contribution	0.0	parallel, non-blocking

5. Bounded Update Law

Bounded Update Law (slide equation)

$$\|\theta_{k+1} - \theta_k\| \leq \delta_{\max} \quad (2)$$

This constraint clips the EKF step size at each cycle. If the raw Kalman update would produce a step larger than $\delta_{\max}$, the update is scaled back to exactly $\delta_{\max}$ in the ℓ_2 norm.

Why this is needed — the adversarial data threat

A **CT-injection attack** (or a CT saturation transient) can cause a sudden large spike in the measured currents, which would cause the EKF innovation $z_k - h(\hat{\theta}_{k|k-1})$ to become very large. Without the bound, a single bad measurement could:

- Shift $\hat{k}_{ibr}$ by 0.5 pu in one cycle, disabling the 87L element.
- Cause the distance relay to “see” a phantom fault location outside Zone 1.

The bounded update law limits the damage any single measurement can do to the state estimate. This connects directly to Slide 17 (Ch 11 cybersecurity) where the adversarial PINN robustness framework is formally established.

Design of $\delta_{\max}$

$\delta_{\max}$ is set to be:

- *Large enough* to track genuine fast fleet changes (typical IBR dispatch ramp: $\Delta k_{ibr} \approx 0.02 \text{ pu/min} = 3.3 \times 10^{-4} \text{ pu/cycle}$ at 50 Hz).
- *Small enough* to reject adversarial step injections ($\delta_{\text{attack}} \sim 0.1\text{--}0.5 \text{ pu/cycle}$ from CT saturation).
- In practice: $\delta_{\max} = 0.05 \text{ pu}$ provides a $150\times$ margin between tracking need and attack rejection.

6. Results — What the Table in the Slide Means

EKF Digital Twin Results Table Explained

Parameter	RMSE	Explanation
$\hat{k}_{ibr}$	0.024 pu	Primary result. On a typical $k_{ibr} = 0.20$ pu fleet, this is 12% relative error — well within the $\pm 10\%$ protection margin. Maintained even at $> 5\%$ PMU measurement noise.
$\hat{Z}_{line}$ (mag.)	0.018 pu	Line impedance magnitude. Lower RMSE than k_{ibr} because Z_{line} changes slowly (temperature drift only) and pre-fault warm-starting provides a very accurate prior.
$\hat{Y}_{shunt}$	0.031 pu	Shunt admittance (charging current). Hardest to estimate — it appears only weakly in the current ratio measurement. Acceptable because 87L stability uses a 3σ threshold that accommodates 0.03 pu Y_{shunt} uncertainty.
Convergence time	<2 cycles	With warm-starting: EKF reaches $< 3\%$ error within 1.9 updates (≈ 38 ms at 50 Hz). This is before Zone-1 distance relay operates (20 ms) only with the warm-start advantage; cold-start would take 13 updates.
Max update latency	1.8 ms	Total DT cycle computation time within the relay firmware. This is the worst-case time the EKF thread holds the shadow register write lock — the relay thread is never blocked for longer than this.

The headline observation

“RMSE ≤ 0.024 pu for k_{ibr} across all operating conditions including $> 5\%$ PMU measurement noise.”

This is the **robustness claim**: even with severely degraded PMU measurements (5% noise is double the IEC class 1P CT error of 2.5%), the DT maintains sufficient accuracy for all downstream protection decisions. This directly addresses a common examiner concern: *“what happens when PMU data quality is poor?”*

7. Best Supporting Figure

Recommended Figure — TR-45 Figure 3 (Non-Blocking Timeline)

Source: SAMBP TR-45/2026, Figure 3 (page 3)

Title: End-to-end fault clearance latency chain

The figure is a horizontal timeline showing:

- **Relay chain** (top, orange): CT/VT (0.5 ms) → MU (1 ms) → relay algorithm (18 ms) → GOOSE (2 ms) → CB (30 ms) = 52 ms total
- **DT thread** (bottom, green): 1 ms cycle running in parallel, *contributing zero latency*
- $T_{\text{mem}} = 100$ ms boundary and 48 ms margin clearly marked

Why it is the best match: The figure directly proves the slide’s central claim — “Update runs in a parallel thread; relay never waits.” It is self-explanatory, has immediate visual impact, and pre-empts the most obvious committee question (“*does the digital twin slow down your relay?*”).

Secondary figure (TR-43, Figure 1): The 4-layer DT architecture diagram (Physical → Relay → Digital Twin → Validation) provides context for explaining the DT’s role as a system-wide parameter hub.

Tertiary figure (TR-44, Figure 3): The DT cycle budget pie chart (285 μ s used, 71.5% margin) explains the 1.8 ms max update latency result and demonstrates computational headroom for future extensions.

8. Connections Across the Thesis

Chapter/Contribution	Connection
C1 — Differential (Ch 3)	$\hat{k}_{\text{ibr}}$ drives slope k_m^* and k_T^* ; $\hat{Y}_{\text{shunt}}$ sets charging current threshold I_c
C2 — OC/Distance (Ch 4)	$\hat{k}_{\text{ibr}}$ gates confidence score γ ; $\hat{Z}_{\text{line}}$ corrects distance reach
C2 — Microgrid (Ch 5)	$\hat{k}_{\text{ibr}}$ feeds the Bayesian mode detector prior (slide 12 equation)
C3 — PINN (Ch 6, next slides)	$\hat{k}_{\text{ibr}}$ is the most important input feature (37 pp permutation importance drop)
C4 — WAPC (Ch 7)	CUSUM monitors the EKF’s $\hat{k}_{\text{ibr}}$ trajectory to detect fleet composition drift and trigger settings update
Appendix A	EKF Jacobian derivation and local observability proof

9. Equations Quick Reference

Key Equations for This Slide

EKF prediction step:

$$\hat{\theta}_{k|k-1} = \hat{\theta}_{k-1|k-1}, \quad P_{k|k-1} = P_{k-1|k-1} + Q \quad (3)$$

Kalman gain:

$$K_k = P_{k|k-1} H_k^\top (H_k P_{k|k-1} H_k^\top + R)^{-1} \quad (4)$$

EKF correction step (slide):

$$\hat{\boldsymbol{\theta}}_{k|k} = \hat{\boldsymbol{\theta}}_{k|k-1} + K_k (z_k - h(\hat{\boldsymbol{\theta}}_{k|k-1})) \quad (5)$$

Bounded update constraint (slide):

$$\|\boldsymbol{\theta}_{k+1} - \boldsymbol{\theta}_k\| \leq \delta_{\max} \quad (\delta_{\max} = 0.05 \text{ pu in deployment}) \quad (6)$$

Cramér–Rao Lower Bound (accuracy floor):

$$\text{RMSE}(\hat{\boldsymbol{\theta}}) \geq \sqrt{\text{diag}[\mathbf{I}(\boldsymbol{\theta})^{-1}]} \quad (7)$$

where $\mathbf{I}(\boldsymbol{\theta})$ is the Fisher information matrix. The reported RMSE values (0.024, 0.018, 0.031 pu) are all within a factor of $1.5\times$ of the CRLB, confirming near-optimal estimation efficiency.

Warm-start convergence speedup (TR-44):

$$\frac{N_{\text{cold}}}{N_{\text{warm}}} = \frac{13.3 \text{ updates}}{1.9 \text{ updates}} \approx 7\times \quad (8)$$

10. TSA Meeting Speaker Script

What to Say at the TSA Meeting

“Slide 13 introduces what I think of as the **nervous system** of the SAMBP framework — the non-blocking EKF digital twin.

Every adaptive threshold I showed in the previous slides — the 87L slope, the OC pickup, the Bayesian mode detector, the distance reach correction — they all depend on knowing k_{ibr} , the current IBR penetration level. The digital twin estimates this continuously from PMU and IED measurements, and broadcasts it to all modules. The estimation uses an Extended Kalman Filter because the measurement model is nonlinear. The filter jointly tracks three parameters: IBR penetration k_{ibr} , line impedance Z_{line} , and shunt admittance Y_{shunt} .

The most important design decision is that the EKF runs in a **parallel thread**. The relay never waits for the EKF to finish. If the filter hasn’t converged after three cycles, the relay simply uses the last valid estimate. This is the shadow register pattern — the relay reads a snapshot; the EKF writes to the background copy without ever blocking the protection logic.

I also impose a **bounded update law**: no single EKF step can change the estimate by more than $\delta_{\max}$. This prevents a CT injection attack or a CT saturation transient from corrupting the state estimate in one cycle and causing a relay mis-operation. The results: RMSE of 0.024 pu for k_{ibr} , 0.018 pu for Z_{line} , and 0.031 pu for Y_{shunt} . These are maintained even with PMU noise above 5% — double the rated class accuracy. Convergence takes less than two power frequency cycles thanks to the pre-fault warm-start. And the maximum EKF thread latency is 1.8 milliseconds,

which contributes **zero additional delay** to the 52-millisecond fault clearance chain.”

11. Anticipated TSA Committee Questions and Answers

1. **Q: What happens if the EKF diverges during a fault?**

A: Divergence is detected by monitoring the innovation sequence. If $\|z_k - h(\hat{\theta}_{k|k-1})\|$ exceeds 3σ for two consecutive cycles, the EKF is re-initialised with the warm-start prior. The relay continues with the last valid estimate throughout. The bounded update law also limits how far a diverging estimate can drift.

2. **Q: The RMSE for Z_{line} is 0.018 pu — how does this affect distance protection accuracy?**

A: The distance Zone 1 reach is set at 80% of the line impedance with a $\pm 10\%$ margin. A 0.018 pu RMSE on a typical $Z_{\text{line}} = 0.05$ pu line is 36% relative error — this *is* significant for impedance magnitude, but the product αZ_{line} (fault location $\times$ impedance) is better conditioned and the distance relay uses the corrected apparent impedance locus, not Z_{line} directly.

3. **Q: Why not use a particle filter or unscented KF instead of EKF?**

A: The EKF Jacobian is analytically derived (Appendix A) and the nonlinearity is mild for $k_{\text{ibr}} \in [0.03, 0.85]$ pu. The EKF update completes in $17.6 \mu\text{s}$; a particle filter with $N = 100$ particles would require ~ 1.76 ms — exceeding the 1.8 ms cycle budget. The UKF is a reasonable future extension but provides no significant accuracy gain for this problem.

4. **Q: How do you handle PMU data dropouts or communication latency?**

A: The EKF prediction step runs even without a new measurement, using only the process model. If a PMU frame is missing, $\hat{\theta}_{k|k} = \hat{\theta}_{k|k-1}$ (no correction applied). This is standard KF missing-data handling and is safe for up to 200 ms (10 missed cycles at 50 Hz) before the estimate uncertainty grows beyond the protection margin.

5. **Q: Is the bounded update law compatible with fast fleet changes, such as sudden IBR disconnection?**

A: A sudden disconnection causes k_{ibr} to drop by up to 0.3 pu in one event. With $\delta_{\text{max}} = 0.05$ pu/cycle, the EKF takes $0.30/0.05 = 6$ cycles (≈ 120 ms) to track this change. During this window, the relay uses a slightly over-estimated k_{ibr} , making protection elements more conservative (higher threshold). This is fail-safe: it produces fewer false trips rather than missed trips.

Study note prepared from SAMBP Technical Reports TR-43, TR-44, TR-45 (2026).

Best supporting figure: TR-45/2026, Figure 3 — End-to-End Non-Blocking Latency Timeline.

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